The Power of Noise: Redefining Retrieval for RAG Systems

Cuconasu, Trappolini, Siciliano, Filice, Campagnano, Maarek, Tonellotto, Silvestri

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Master AI — Information Retrieval 1

Presented by Chris Hoes & Pepijn van der Klei

Problem

Limitations of LLM's:

- Knowledge is frozen at training time
- Prone to hallucinations on factual questions
- Struggle with domain-specific or rapidly changing knowledge

> Asking an LLM about the weather today?

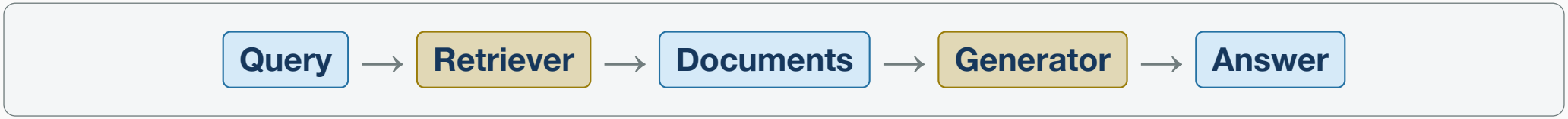
Solution?

How can we make LLMs access up-to-date knowledge?

Solution!

Retrieval-Augmented Generation (RAG):

- Use an IR system to retrieve relevant documents for the query
- Feed these documents into the LLM prompt as context



But?

Common assumption:

- Feed top-k highest-scoring retrieved docs to the LLM
- More semantically relevant == better
- Our current retrieval systems are good at this

But: the retriever's role itself has received surprisingly little systematic study.

Not So Fast

These assumptions are not actually true:

- ~~Feed top-k highest-scoring retrieved docs to the LLM~~
- ~~More semantically relevant == better~~
- ~~Our current retrieval systems are good at this~~

> We already discussed this in class — and this paper was referenced as evidence that common retrieval assumptions break down in the RAG setting.

So what **does** make a good retriever for RAG? Let's look at the study.

Research Question

> *"What characteristics are desirable in a retriever to optimize prompt construction for RAG systems? Are current retrievers ideal?"*
>
> Three dimensions studied: **type**, **position**, and **number** of documents.

Contributions:

1. First comprehensive study on how the *type* of retrieved documents affects LLM performance in RAG
2. New retrieval heuristics based on the (unexpected) results
3. All code and data publicly released

A Taxonomy of Retrieved Documents

The paper formalises four document types a retriever can return:

★ Gold — the **ground-truth passage** containing the answer.

● Relevant — other passages that **also contain the answer** (not directly tested).

▲ Distracting — semantically close to the query but **no answer** (top-k but wrong).

■ Random/Noise — **unrelated** to the query; pure noise.

Prompt structure notation:

I	Task instruction
★	Gold document
●	Relevant documents
▲	Distracting documents
■	Random documents
Q	The query

Example: `[I, ▲, ★, Q]` means: instruction, then distracting docs, then gold, then query.

The **position** of each document relative to Q will matter greatly.

Experimental Setup

📄 Dataset

NQ-open (Natural Questions)

- 21M Wikipedia passages (100 words each)
- 2,889 test queries
- Answers ≤ 5 tokens
- Open-domain Q&A

🤖 Generators

- **Llama2-7b** (main model)
- Phi-2
- Falcon-7b
- Mosaic MPT-7b

All open-source, 4-bit quantised
Greedy decoding, max 15 tokens

🔍 Retrievers

- **Contriever** (dense, bi-encoder)
- **BM25** (sparse, term-based)

Dense vs. sparse comparison

Metric: Accuracy — at least one reference answer must appear in the generated response

Significance: Wilcoxon signed-rank test ($p < 0.01$) throughout

Finding 1: Position of the Gold Document Matters

Three positions tested for the gold document ⋆ Gold relative to the query **Q**:

Near: [I, ▲, ⋆, Q]
Gold adjacent to query

Far: [I, ⋆, ▲, Q]
Gold at start of prompt

Mid: [I, ▲, ⋆, ▲, Q]
Gold buried in the middle

Near > Far > Mid. The middle of the context is hardest to attend to — worse than the start of the prompt.

Intuition from attention patterns:

- Attention scores are **highest near the query** — and also relatively high at the very start of the prompt
- The **middle** of the context gets the least attention: a U-shaped curve
- This is exactly the **"Lost in the Middle"** effect (Liu et al., 2023)

Practical implication: place relevant docs **immediately before** the query. Start of prompt is second best — never bury them in the middle.

Consistent across all tested models (Llama2, Phi-2, Falcon, MPT).

Finding 2: Distracting Documents

Common perception:

- Retrieve top-k documents by similarity score
- Semantically close documents are helpful for the LLM
- More top-scoring documents → better answers

> In practice, retrievers often return documents that are **semantically aligned** with the query but **do not contain the answer**. These are called **distracting documents** ▲ Distracting.

Distracting Documents — Example

Query: *"Who owned the Millennium Falcon before Han Solo?"*

▲ Distracting **Doc 1:**

Before the events of the film, he and Chewbacca had lost the "Millennium Falcon" to thieves, but they reclaim the ship after it…

▲ Distracting **Doc 2:**

The "Falcon" has been depicted many times in the franchise, and ownership has changed several times…

⋆ Gold **Doc 3:**

Han Solo won the Millennium Falcon from **Lando Calrissian** in the card game sabacc…

Answer: Han Solo (correct: Lando Calrissian)

Distracting Documents — Experiment

Sub-question:

> How do distracting documents ▲ **Distracting** affect RAG performance?

Setup:

- Oracle setting: the gold document ⋆ **Gold** is always in the prompt
- Add 0–10 distracting documents (high retrieval score, no answer)
- Test across all four LLMs and three positions (far, mid, near)
- Schematically: `[I, ▲..., ⋆, Q]`

Baseline comparison (Llama2, Contriever, Near):

Only gold `[I, ⋆, Q]`	**0.564**
Gold + 1 distracting `[I, ▲, ⋆, Q]`	↓ **0.428**
Gold + 2 distracting `[I, ▲▲, ⋆, Q]`	↓↓ **0.397**

Distracting documents **▲ Distracting** (high retrieval score, **no answer**) **decrease** LLM accuracy vs. using only the gold document.

The **retriever's ranking score is not a reliable signal** for downstream LLM usefulness. High retrieval score ≠ helpful in the prompt.

Distracting Documents — Results

	Far - [I, ★, ✿, Q]				Mid - [I, ✿, ★, ✿, Q]				Near - [I, ✿, ★, Q]			
# ✿	Llama2	MPT	Phi-2	Falcon	Llama2	MPT	Phi-2	Falcon	Llama2	MPT	Phi-2	Falcon
0	**0.5642**	0.2148	**0.4438**	**0.4330**	**0.5642**	**0.2148**	**0.4438**	**0.4330**	**0.5642**	**0.2148**	**0.4438**	**0.4330**
1	0.4586	0.1976	0.3585	0.3469	no-mid	no-mid	no-mid	no-mid	0.4283	0.1791	0.4227	0.3602
2	0.3455	0.1913	0.3430	0.3246	0.3322	0.1802	0.3375	0.2823	0.3974	0.2002	0.3975	0.3111
4	0.2745	**0.2209***	0.3019	0.2670	0.2857	0.1775	0.2885	0.2378	0.3795	0.2059*	0.3701	0.2736
6	0.2898	0.2171*	0.2943	0.2392	0.2698	0.1424	0.2625	0.2103	0.3880	0.1892	0.3623	0.2656
8	0.2643	0.2077*	0.2513	0.1878	0.2268	0.1002	0.2360	0.1745	0.3748	0.1944	0.3423	0.2424
10	0.2537	-	-	-	0.2180	-	-	-	0.3716	-	-	-
12	0.2688	-	-	-	0.2382	-	-	-	0.3991	-	-	-
14	0.2583	-	-	-	0.2280	-	-	-	0.4118	-	-	-
16	0.2413	-	-	-	0.2024	-	-	-	0.3889	-	-	-
18	0.2348	-	-	-	0.1795	-	-	-	0.3781	-	-	-

- Progressive degradation across **all LLMs** — accuracy drops up to 67%
- Even **one** distracting document causes up to 25% accuracy loss
- Consistent across all three positions (far, mid, near)

- Confirmed with **ADORE** (hard-negative retriever) — same degradation
- Distinguishing relevant from distracting is a hard problem that **cannot be mitigated** by changing the retrieval method

Distracting Documents — Explanation

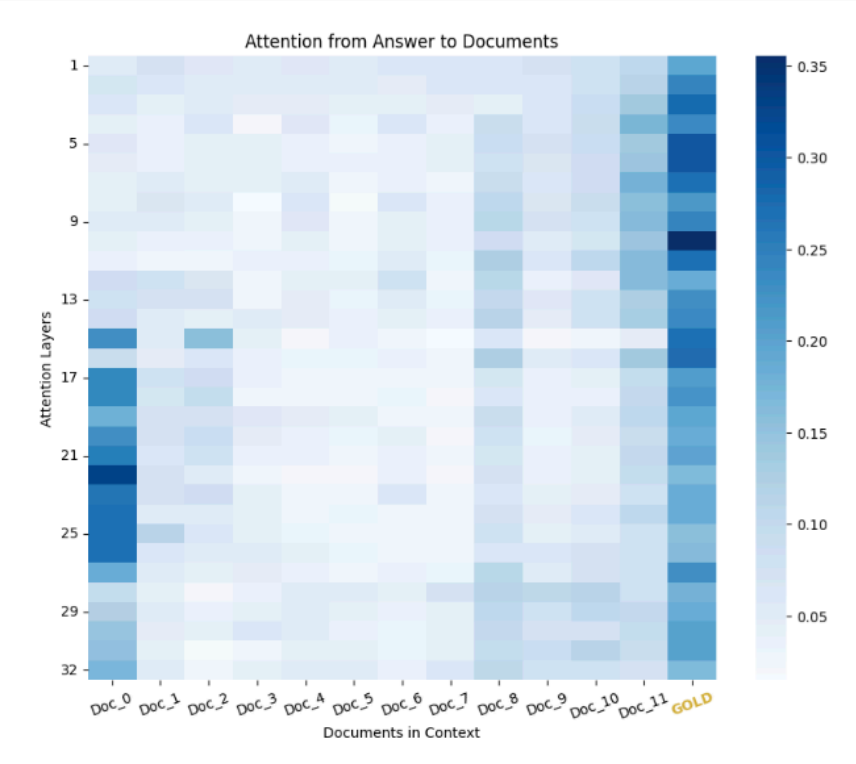


Figure 3: Attention from answer tokens to context documents in [I, ▲..., ⋆, Q] (Llama2).

The paper's attention analysis shows the LLM **disproportionately attends** to distracting documents over the gold document, explaining the wrong answers.

> Semantically aligned documents compete for the model's attention — the LLM cannot distinguish between **relevant-with-answer** and **relevant-without-answer**.

> **Note:** the most-attended document is Doc_0 (first in prompt) and the gold is Doc_12 (last, near the query). Is this a property of distracting docs, or does position itself drive attention? LLMs are known to attend more to the start and end of the context.

Finding 3: Random Noise *Improves* Accuracy — Oracle Setting

Oracle setting: the gold document is always included. Random Wikipedia passages are added around it.

Prompt: `[I, ▇▇▇, ⋆, Q]` — noise + gold near Q

Results (MPT-7b, Near):

Only gold `[I, ⋆, Q]`	0.215
1. 4 random docs (best)	**0.293**

+36% accuracy improvement from adding documents that contain no answer at all!

Not consistent across all models:

- **MPT:** noise helps in *all* positions (Near, Far, Mid)
- **Llama2 & Phi-2:** noise helps only in the **Near** setting; hurts in Far/Mid
- **Falcon:** noise does *not* improve accuracy in this oracle setting

The benefit of noise is **model-dependent** in the oracle setting — but Near positioning + noise is the most robust combination.

Best noise position: far from query; gold near query. `[I, ▇▇▇, ⋆, Q]` outperforms `[I, ⋆, ▇▇▇, Q]`.

Finding 4: Randomness in Practice

So far — oracle setting:

- Gold document ⋆ Gold is **always** present in the prompt
- We control exactly which documents are added
- Not representative of a real RAG pipeline

In practice:

- You retrieve top-k documents — a mix of relevant, distracting, and possibly gold
- No guarantee the gold document is even retrieved
- You don't know which document is which

> Can random noise still improve accuracy when the prompt contains **real retrieved documents** instead of a hand-picked gold document?

Sub-question:

> Does padding retrieved documents with random noise improve accuracy in a realistic end-to-end RAG setting?

Setup:

- Retrieve top-k documents using an IR system (no oracle — the retrieved set is a mix of • Relevant, ▲ Distracting, and possibly ⋆ Gold)
- Pad remaining context window with random documents ■ Random/Noise
- Prompt: [I, ■■■, ▲/●/⋆, Q] — noise **before** retrieved docs
- Tested on **Llama2** (Table 3) and **Falcon** (Table 5)
- Both **Contriever** (dense) and **BM25** (sparse) retrievers
- Additional noise types: **Reddit** posts and **random words** (nonsensical sentences)
- Evaluated on NQ-open **test set** (earlier experiments used training set)

Randomness in Practice — Results

Noise still works — even better:

- Adding noise improves accuracy up to **+35%** (Contriever, 4 retrieved docs) and **+44%** (BM25)
- Almost always beneficial regardless of number of retrieved docs
- Filling the context window with noise until full yields best results

4 retrieved only (Contriever)	0.187
1. fill with noise (Contriever)	**0.253** **(+35%)**
4 retrieved only (BM25)	0.203
1. fill with noise (BM25)	**0.293** **(+44%)**

Broader than expected:

- **Falcon** now benefits from noise — unlike in the oracle setting where it showed no improvement
- Holds for **BM25** too — BM25 shows 3–4 pp higher accuracy on average (higher top-k accuracy)
- **Noise source doesn't matter**: Reddit posts and random words work at least as well — Reddit even adds +9% over Wikipedia noise

Sweet spot: retrieve **3–5 docs**, pad remaining context with noise. More docs → more distractors.

Randomness in Practice — Detailed Results

Table 3 — Llama2: Wikipedia noise + retrieved docs (Contriever vs BM25)

	Contriever							BM25						
# 📄 / # ⋮⋮	1	2	3	4	5	8	10	1	2	3	4	5	8	10
0	0.1620	0.1866	0.1876	0.1866	0.1921	0.2198	0.2108	0.2008	0.2208	0.2084	0.2028	0.2243	0.2492	0.2447
1	0.1308	0.1616	0.1717	0.1893*	0.1987*	0.2153*	0.2146*	0.1568	0.1963	0.1921	0.2115	0.2295*	0.2475*	0.2506*
2	0.1315	0.1644	0.1859*	0.2008	0.2174	0.2156*	0.2368	0.1644	0.1973	0.2080*	0.2281	0.2558	0.2495*	0.2596
3	0.1301	0.1727	0.2008	0.2316	0.2201	0.2198	0.2409	0.1568	0.2063	0.2160	0.2520	0.2579	0.2644	0.2707
5	0.1464	0.2056	0.2233	0.2240	0.2150	**0.2451**	**0.2482**	0.1772	0.2402	0.2437	0.2520	0.2554	0.2804	**0.2866**
8	0.1734	0.2066	0.2336	0.2375	0.2454	0.2416	0.2364	0.1994	0.2451	0.2579	0.2769	0.2817	**0.2859**	0.2777
10	0.1796	0.2174	0.2450	0.2502	**0.2499**	0.2420	-	0.2108	0.2589	0.2734	0.2835	**0.2935**	0.2853	-
15	0.2018	0.2354	0.2551	**0.2530**	-	-	-	0.2243	0.2686	0.2790	**0.2928**	-	-	-
16	0.2032	**0.2471**	**0.2558**	-	-	-	-	0.2323	0.2662	**0.2838**	-	-	-	-
17	0.2039	0.2426	-	-	-	-	-	**0.2326**	**0.2693**	-	-	-	-	-
18	**0.2073**	-	-	-	-	-	-	0.2309	-	-	-	-	-	-

Rows = random docs added, columns = retrieved docs. Bold = best per column. (*) = not significant vs. baseline.

Randomness in Practice — Detailed Results (Falcon)

Table 5 — Falcon-7b: Reddit noise + retrieved docs [I, ▬▬, ▲/●/★, Q]

# ▦ \ # 🗎	1	2	3	4	5	9
0	0.1568	0.1717	0.1855	0.1938	0.1942	**0.1998**
1	0.1551*	0.1793*	0.1897*	0.1924*	0.1976*	-
2	0.1529*	0.1762*	0.1938*	0.2011*	0.1976*	-
3	0.1599*	0.1727*	0.1911*	0.2021*	**0.2118**	-
4	0.1606*	0.1758*	0.1959	0.2073	0.2108	-
5	0.1627*	0.1762*	0.2000	**0.2108**	-	-
6	0.1651*	0.1848	**0.2004**	-	-	-
7	0.1675	**0.1848**	-	-	-	-
8	**0.1682**	-	-	-	-	-

- Falcon **did not** benefit from noise in the oracle setting
- In the realistic setting, noise **does** improve Falcon — consistently across all configurations
- Confirms that the noise benefit generalises beyond the oracle experiments

> The practical benefit of noise is **more general** than oracle results suggested — it extends to models that previously seemed unresponsive to noise.

Why Does Noise Help? Entropy Collapse

From prior work (Attanasio et al., 2022; Hoffmann et al., 2023):

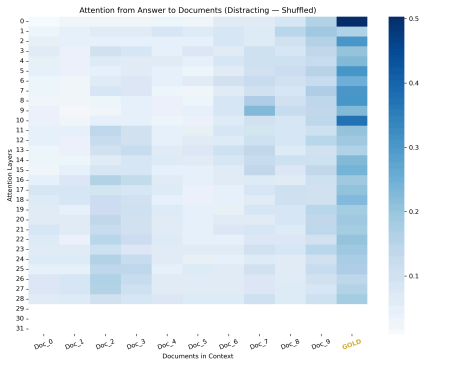
- There are cases where pathologically **low attention entropy** causes LLMs to generate degenerate outputs with a sharp decrease in performance
- These episodes are called **entropy collapse**
- Low entropy = the model focuses too narrowly on a small region of the context, ignoring relevant information

The authors' hypothesis:

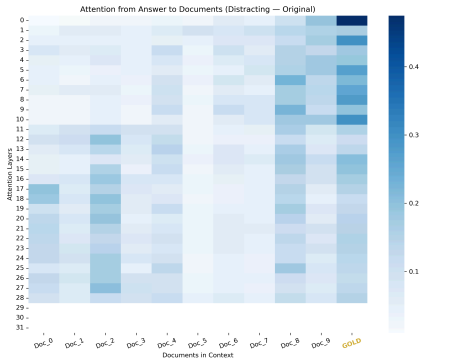
> By adding random documents to the context, we are **better conditioning** the generative function $p_\theta(y \mid \cdot, d)$, inducing enhanced accuracy. Random documents may act as "attention anchors" that **prevent entropy collapse** — spreading the model's attention more evenly and reducing degenerate focus on irrelevant passages.

Entropy Collapse — Evidence

Lower entropy:



Higher entropy:



Attention entropy experiment:

Measured on **Llama2-7b** in the oracle setting.

Compare the attention entropy of:

- Only gold: `[I, ★, Q]`
- With noise: `[I, ■■, ★, Q]`

Attention entropy **increases 3x** when random documents are added to the prompt.

Higher entropy = attention is more **distributed** across documents, rather than collapsing onto a narrow, potentially misleading region.

Entropy Collapse — Open Question

The entropy collapse hypothesis shows **correlation** (noise → higher entropy → better accuracy) but **not causation**. The authors explicitly note: *"although these experiments show a pattern, we cannot yet answer this question in a definitive manner."*

- Investigating **why** LLMs respond this way to noise is outside the scope of the paper
- The paper focuses on the **retriever** component — not the inner workings of the LLM
- Flagged as an important direction for **future work**: understanding what characteristics of noise make it advantageous

Practical Heuristics from the Paper

Proposed retrieval strategy:

- Retrieve **3–5 documents** with your IR system
- Place them **near the query** in the prompt
- Fill the remaining context with **random documents**
- Put random filler **far from the query**

Result: `[I, ■ ■ ■, ▲/●, Q]`

Why not retrieve more?

More retrieved docs → increasing chance of **distracting** content → accuracy degrades.

The sweet spot for the studied setup: $k = 3$ **to** 5.

Key insight for the field:

Retrieval quality for RAG is **not** the same as classic IR relevance ranking. A retriever optimised purely for top-k relevance may **hurt** the downstream LLM. New retrieval objectives are needed.

Connecting to the Course

Course concepts in this paper:

RAG (Week 4)
The RAG lecture introduced this exact problem: semi-relevant docs as distractors and Lost in the Middle. This paper provides the empirical deep-dive.

Dense retrieval — Contriever (Week 3)
Bi-encoder from the Dense Retrieval lecture; used as the main dense retriever in the paper.

BM25 (Weeks 1 & 3)
TF-IDF weighting (Week 1) → BM25 as sparse retriever (LSR lecture). Paper compares sparse vs. dense for RAG.

Recall@k (Week 2)
Standard metric from the Evaluation lecture — but the paper shows high Recall@k does **not** guarantee good LLM accuracy.

What this paper adds beyond the course:

> Classic IR optimises for **document relevance to query**. RAG requires optimising for **LLM generation quality given the prompt**. These are **not the same objective**.

The Week 4 RAG lecture flags distractors and Lost in the Middle as open challenges. This paper **quantifies** both — and reveals the surprising noise effect not discussed in the course.

The paper's call for new retrieval objectives aligns with the course's progression: classical ranking (Weeks 1–2) → neural retrieval (Week 3) → generation-aware retrieval (Week 4).

Conclusions

Three key findings — remember these:

- **Position matters.** Place relevant docs **near the query** — the LLM cannot attend well to distant context.
- **Distracting docs hurt.** Non-relevant top-k docs **reduce** accuracy.
- **Noise helps.** Random docs **improve** accuracy by up to 35–44%.

Published at SIGIR 2024. Code & data released at
`github.com/florin-git/The-Power-of-Noise`
129 citations as of Feb 2026.

Open questions & future work:

- **Why** does noise help? The entropy hypothesis is suggestive but not proven.
- Which properties make a document useful as filler?
- Can we **train** a retriever with an objective aligned to LLM accuracy, not just relevance?
- Do these findings hold for larger LLMs (13B, 70B)?
- How do results change with instruction-tuned / RLHF models?